Non-Destructive Root Volume Estimation

Project G14, Crop Breeding Trials. A project report.

Summary

This project asks a simple question with a hard answer: can we work out how much root a plant has without digging it up? Measuring root volume the traditional way means uprooting the plant, which destroys the very thing you want to track over time. We tried to get around that by training deep-learning models on underground scan images from the CGIAR Root Volume Estimation Challenge on Zindi.

The honest headline is that the problem is difficult and our models have not solved it yet. Across five-fold grouped cross-validation, every approach we tried, including a plain image CNN, a metadata CNN, a multimodal CNN and simple ridge baselines, landed at almost the same error (RMSE around 1.47 to 1.52) and explained very little of the variation in root volume (R^2 between 0.00 and 0.07). The best result came from a deliberately simple baseline, not a neural network. What we did build is a solid, reproducible pipeline, a careful evaluation setup, and a working approach to uncertainty estimation, which gives us a firm base for the next stage.

Why this matters

Root volume tells breeders a lot: how well a plant can pull in water and nutrients, and how it might cope with stress. For crop breeding trials, that information is valuable, but getting it usually means destructive sampling. You cannot measure the same plant twice if you have to pull it out of the ground the first time. A reliable image-based estimate would let researchers follow root development over time and compare genotypes without sacrificing the plants.

Just as important is knowing how much to trust a prediction. A number like "3.2" is much less useful than "3.2, probably between 1.0 and 5.4". For that reason, uncertainty was a goal of the project from the start, not an afterthought.

The data

We used the CGIAR Root Volume Estimation dataset from Zindi. The training set has 386 labelled observations spread across 98 folders. Each observation comes with metadata (folder name, plant number, side, start and end positions, genotype and growth stage) and a root-volume label. The test set has the same structure but no labels.

The scans themselves are grayscale images stored in RGBA format. A single observation is not one picture but several image strips, which belong together in slice order.

Two features of this dataset shaped almost everything that followed:

- **It is small.** 386 labelled observations is not much for a deep-learning model, especially when the images are complex.
- **The labels are lopsided.** Most plants have low root volumes, and the high end is thinly covered.

Root-volume range	Observations
0 to 1	112
1 to 2	93
2 to 3	112
3 to 4	34
4 to 5	21
Above 5	14

Only 3.63% of the training observations have a root volume above 5, so the models see very few examples of large roots.

We also noticed something awkward during the data audit: identical image sequences sometimes appear with different plant numbers and different root-volume labels. If two identical inputs have different answers, there is a ceiling on how well any model can do, and that is worth keeping in mind when reading the results.

What we built

Preparing the images

For each observation we loaded its image strips, converted them to grayscale and stacked them vertically. We then resized the stack to 512 pixels wide (keeping the aspect ratio) and padded it to a height of 256 pixels. Each input has two channels: the grayscale intensity, and a validity mask that marks which pixels are real image and which are padding. The mask lets the model tell the difference between a dark region of soil and an empty region we added ourselves. We did not apply per-image intensity normalization.

The models

The image encoder is a small convolutional network with channel widths of 2, 16, 32, 64 and 128, followed by adaptive average pooling and a regression head. We built three versions of the model so we could see where the useful signal was coming from:

- an **image-only CNN**,
- a **metadata-only CNN**, using plant number, genotype, stage, side, start, end and strip count,
- a **multimodal CNN** that combines both.

We trained with AdamW (learning rate 0.0003, weight decay 0.0001), a batch size of 8 and early stopping for the first experiments. Our main loss was SmoothL1, and we also tried plain MSE and a log-target version.

Evaluating fairly

Because one folder contains several related observations, a random split would let the model peek at near-copies of its test data and flatter the results. We avoided this by using five-fold cross-validation grouped by folder, so every folder is held out as a whole. We report RMSE, MAE and R^2, and we also built baselines that do not use a neural network at all: ridge regression on image area, on metadata, and on both together. If a deep model cannot beat a simple ridge, that is important to know.

What we found

Cross-validation results

These are pooled out-of-fold results from the five grouped folds.

Model	RMSE	MAE	R^2
Image CNN	1.4981	1.0765	0.0342
Metadata CNN	1.4769	1.0371	0.0612
Multimodal CNN (SmoothL1)	1.5240	1.0745	0.0004
Multimodal CNN (MSE)	1.4985	1.0682	0.0336
Multimodal CNN (log-MSE)	1.5160	1.0572	0.0109

Baseline	RMSE	MAE	R^2
Image-area Ridge	1.5010	1.0650	0.0304
Metadata Ridge	1.4861	1.0791	0.0495
Metadata + image-area Ridge	1.4728	1.0650	0.0665

The most striking thing about these tables is how similar everything looks. The best model overall, the metadata plus image-area ridge, has an RMSE of 1.4728, and the worst CNN is at 1.5240. That is a small spread, and with data this limited we would not read much into the ordering. R^2 values near zero mean that none of these models explains much more variation than simply predicting an average root volume for every plant.

Two takeaways stand out. First, the CNNs did not beat the simple baselines, so at this stage the extra complexity is not paying for itself. Second, the multimodal model did not improve on its single-input cousins, which suggests that the model is not yet extracting complementary information from the images and the metadata.

Where the models go wrong

Both the image and metadata CNNs show a clear pattern: they underestimate high root volumes and overestimate very low ones. That is what you would expect when most training examples sit in the middle and few sit at the extremes. The models play it safe and pull their predictions toward the average. Given that only 3.63% of observations are above 5, it is not surprising that large roots are the hardest to predict.

Performance also varied from folder to folder, so some groups of plants are simply harder than others.

Uncertainty estimation

To give predictions an honest range, we used split conformal prediction with folder-disjoint training, calibration and evaluation sets. Using the image CNN, we got these results on a separate holdout experiment:

- Evaluation RMSE of 1.4848, MAE of 1.1963 and R^2 of -0.0705.
- Observation-level empirical coverage of 92.68%, against a 90% target.
- A mean interval width of 4.46.

In plain terms, the intervals did cover the true value slightly more often than promised at the level of individual observations. But an interval of 4.46 units wide is very wide when most root volumes fall between 0 and 5, so the intervals are honest but not especially useful yet. Their width reflects how uncertain the underlying model really is.

When we checked coverage at the level of whole folders, using a folder-maximum calibration method, we reached 85% on 20 evaluation folders, which is under the 90% target. That tells us the method is less reliable when we care about getting every observation in a folder right at once.

A caution on reading these numbers: they come from a separate holdout experiment and should not be compared directly with the cross-validation tables above. Coverage measured on a single split also does not guarantee the same coverage on future data.

Limitations

We want to be upfront about what this project does not show.

- **Small dataset.** With 386 labelled observations, it is hard to train an image model that generalizes, and the results are noisy.
- **Imbalanced targets.** High and very low root volumes are poorly represented, which biases predictions toward the middle.
- **Conflicting labels.** Identical image sequences appear with different plant numbers and different targets, which puts a limit on achievable accuracy.
- **Weak overall signal.** R^2 near zero means the current models are not yet useful for precise prediction.
- **Wide intervals.** The uncertainty ranges are valid-looking but too broad to guide decisions with confidence.
- **Unvalidated application.** The desktop GUI uses an experimental checkpoint and should not be treated as a validated deployment tool.

What comes next

The project is now in the model-improvement and application-development stage. Our plan has three strands.

Better models. We want to attack the imbalance directly, with weighted SmoothL1 and weighted MSE losses that give rare high and low volumes more influence, and to revisit log-target regression. We also want to look into folder-specific and genotype-related effects, and to rethink both the image representation and the CNN architecture.

More trustworthy validation. Single runs on a small dataset can mislead, so we plan to repeat the grouped experiments across several random seeds. We will also measure coverage at both the observation and whole-folder level, check calibration within subgroups, test how stable calibration is across folds, and work on making the intervals tighter without losing coverage.

A usable tool. Once a configuration is selected, we will train a final model, calibrate it on a separate calibration set, and improve the GUI with prediction history, CSV export and better visualisation. We will also document its inference requirements and limits clearly, so that nobody mistakes it for something more reliable than it is.

Conclusion

In this project we set out to estimate root volume without harming the plant, and we built the full pipeline to attempt it: data auditing, preprocessing, three CNN variants, simple baselines, grouped cross-validation, and conformal uncertainty estimates. The results are modest. The models explain little of the variation in root volume, and a simple ridge baseline matched or beat the neural networks.

Still, that is a useful result in its own right. It tells us that the difficulty here lies in the data (its size, its imbalance and its inconsistent labels) at least as much as in the model design. We now have a reliable evaluation setup that will not flatter us, a clear picture of where predictions fail, and a concrete plan to improve them. The end goal is unchanged: a reproducible, non-destructive tool that gives a root-volume estimate together with an uncertainty range that people can trust.